Stochastic Modeling and Simulation

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Stochastic Processes and their Classification
Chapter 1 : Discrete Time Markov Chains

Introduction to stochastic processes

- Definition: A Discrete-Time Markov Chain (DTMC) is a stochastic process where changes happen at discrete time steps.
- Events occur at the end of each time step $\longrightarrow$ synchronized updates.
- DTMCs are not suited for all scenarios but they are well-adapted for modeling systems envolving discrete steps.
- Contrast: Continuous-Time Markov Chains (CTMCs) allow events to happen at any instant, better for continuous-time systems.

Stochastic Processes

- A **stochastic process** is a sequence of random variables indexed by time X_t and defined on the same probabilized space $\{\Omega, \mathcal{A}, \mathbb{P}\}$

$$\{X_t, t \in T\},$$

- **Discrete-time process:** $\{X_n, n \geq 0\}$, with time set T is finite or at least countable (most often $T = \mathbb{Z}_+$).
- **Continuous-time process:** $\{X(t), t \geq 0\}$, with time set T is uncountable (typically $\mathbb{R}_+$).
- **State space** S of the process is the set of all possible values of the random variables.
- If S is finite or countable state space, the proces is called a **chain**.

Markov property

Definition

A **Markov** process $\{X_n, n = 0, 1, 2, ...\}$, the future X_{n+1} depends only on the present state X_n, not on past states:

$$\begin{aligned} & \mathbb{P}\left[X_{n+1} = i_{n+1} / X_n = i_n, X_{n-1} = i_{n-1}, ..., X_0 = i_0\right] \\ = \; & \mathbb{P}\left[X_{n+1} = i_{n+1} / X_n = i_n\right]. \end{aligned} \tag{1}$$

This is the **memoryless property**.

Discrete-Time Markov Chain

Definition

A **Discrete-Time Markov Chain** is a stochastic process $\{X_n, n \geq 0\}$ satisfying:

- The process has a discrete time;
- the states space S is finite or countable;
- the process satisfies the Markov property.

Additional assumptions:
Time-homogeneity: Transition probabilities depend only on the current state, not on time step:
For all instant n, $k \geq 0$ and for any pair (i, j) ,

$$\mathbb{P}\left[X_n = j / X_{n-1} = i\right] = \mathbb{P}\left[X_{n+k} = j / X_{n+k-1} = i\right], \forall k \geq 0. \quad (2)$$

Representative Graphs

A MC or more precisely its transition matrix, can be represented by an oriented graph G whose vertices correspond to the states of the chain and where an edges connects the vertices associated with the states i and j if the transition probability from i to j is positive, the graph thus defined is called the **representative graph** of the Markov chain.

Transition probability and matrix

Definition

The probability p_{ij} is called **probability of transition** (or **of passage**) from the state i to the state j in one-step and is equal to the conditional probability that the system will be in state j at the next step, given that it is currently in state i.

$$\mathbb{P}\left[X_n = j / X_{n-1} = i\right] = p_{ij}. \tag{3}$$

Transition probability and matrix

Definition

A **transition probability matrix** of DTMC with $|S| = n$ states is an $n \times n$ matrix P where:

$$P = (p_{ij}).$$

Transition matrices are also called **stochastic matrices** and satisfy the following two conditions:

$$\forall i, j \in S : 0 \leq p_{ij} \leq 1 \text{ and } \sum_{j \in S} p_{ij} = 1, i \in S.$$

Stochastic matrix, captures full dynamics.

Transition probability and matrix

Example (Repair facility problem)

A machine is either working or in the repair center. If it is working today, then there is a 95% chance that it will be working tomorrow. If it is in the repair center today, then there is a 40% chance that it will be working tomorrow. We are interested in questions like "what fraction of time does my machine spend in the repair shop?"
Question 1. Describe the DTMC for the repair facility problem.
Question 2. Now soppose that after the machine remains broken for 4days, the machine is replaced with a new machine. How does the DTMC diagram change?

Power of P: n-Step Transition Probabilities

To compute probabilities over multiple steps, we define:

$$p_{ij}^{(m)} = \mathbb{P}\left[X_{n+m} = j / X_n = i\right]$$

for the conditional probability of going from i to j in exactly m steps.

The probability of being in state j in m-steps, given that we are in state i now, called **Chapman-Kolmogorov**, is given by

$$p_{ij}^{(m)} = \sum_{k \in S} p_{ik}^{l} p_{kj}^{m-l}, \text{ for any } l < m.$$

Power of P: n-Step Transition Probabilities

In matrix form, this property becomes

$$P^{(m)} = \left(p_{ij}^{(m)}\right) = P^m,$$

$P^m = P.P...P,$ multiplied m times.
The matrix of **m-step transition probabilities** is obtained by the matrix powers. We can use the notation P_{ij}^m to denote $(P^m)_{ij}$.
Used for long term analysis and hitting probabilities.

Example

Repair facility problem: Now we consider again the simple repair facility problem, with general transition probability matrix P :

$$P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}, 0 < a < 1, 0 < b < 1.$$

We prove by induction that

$$\begin{aligned} P^n &= \begin{bmatrix} \frac{b+a(1-a-b)^n}{a+b} & \frac{a-a(1-a-b)^n}{a+b} \\ \frac{b-b(1-a-b)^n}{a+b} & \frac{a+b(1-a-b)^n}{a+b} \end{bmatrix} \\ \lim_{n\to\infty} P^n &= \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}. \end{aligned}$$

Example

let the transition probabiliies matrix associated to a DTMC:

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 \end{pmatrix}, P^5 = \begin{pmatrix} 0.06 & 0.3 & 0.64 \\ 0.18 & 0.38 & 0.44 \\ 0.38 & 0.44 & 0.18 \end{pmatrix},$$

$$P^{30} = \begin{pmatrix} 0.230 & 0.385 & 0.385 \\ 0.230 & 0.385 & 0.385 \\ 0.230 & 0.385 & 0.385 \end{pmatrix}.$$

Observe that all the rows become the same! Note also that, for all the above powers, each row sums to 1.

Classification of Chain States

There are several type of states. The classification of the states of a Markov chain plays an essential role in the study of its long-term behavior.

Accessibility: A state j is **accessible** from state i, denoted $i \to j$, if there exists $l \geq 0$ such that $p^l_{i,j} > 0$.

Communication: State i and j **communicate,** denoted $i \leftrightarrow j$, if i leads to j and j leads to i, if $\exists l, m \in \mathbb{N} : p^l_{i,j} > 0$ and $p^m_{j,i} > 0$.

If $i \leftrightarrow j$, it means there is a non-zero probability of moving from i to j, and vice versa.

This binary relation is an equivalence relation.

Classification of Chain States

Type of States

Definition

Let (X_n) be a Markov chain and let j be a fixed state. For any integer $n \geq 1$

$$f_{ij}^{n} = \mathbb{P}\left[X_n = j, X_k \neq j, k = 1, 2, \cdots, n-1 / X_0 = i\right].$$

is the probability that if the initial state is i, the first transition to state j occurs at the n^{th} transition.

Classification of Chain States

Type of States

- **Recurrent State**

Definition

A state i is **recurrent** if

$$f_{ii} = \mathbb{P}\left(\text{returning to state } i/\text{starting from state } i \text{ (initial)}\right) = 1.$$

Proposition. If i and j communicate and i is recurrent, then j is recurrent.

Classification of Chain States

Type of States

- **Transient State**

Definition

A state i is called **transient** if there is a non-zero probability that the chain will never return to that state once it leaves. Otherwise $(f_{ii} < 1)$, the state is **transient**.

Classification of Chain States
Type of States

We denote the expected return time: m_{ii} which is the **expected number of steps** it take for the chain to return to state i after starting there. It is defined as

$$m_{ii} = E\left[T_i / X_0 = i\right] = \sum_{n=1}^{\infty} n f_{ii}^{(n)}.$$

where T_i is r.v representing the time (or the number of steps required for the chain) of the first return to state i,
$T_i = \inf\left\{n \geq 1, X_n = i\right\}.$

Classification of Chain States

Type of States

- **Positive Recurrent State**

  A state i is **positive recurrent** if the expected return time to that state is finite: $m_{ii} < \infty$.
  The average number of steps it takes to return to state i (starting from i) is a finite number.
  These states are those for which the chain doesn't take "too long" to revisit.

- **Zero Recurrent State**

  A state i is **zero recurrent** (or **null recurrent**) if it is recurrent but the expected return time is infinite, an extremely long time on average (possibly an infinite amount of time), $m_{ii} = \infty$.

Classification of Chain States

Type of States

- **Absorbing States**

Definition

A state i is **absorbing** if $p_{ii} = 1$ and $p_{ij} = 0$ for all $j \neq i$, meaning that once the chain enters this state, it cannot leave.

Classification of Chain States

Type of States

- **Periodic and Aperiodic States**

Definition

The **period** $d(i)$ of a state i is defined as

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{n} > 0\}$$

A state i is **periodic** if $d(i) > 1$.
Otherwise $(d(i) = 1)$, a state is **aperiodic** if it has period 1. A chain (X_n) is said to be aperiodic if all of its states are aperiodic.

Remark

If i and j communicate, so $d(i) = d(j)$.

Classification of Chain States

Type of States

- **Ergodic states**

Definition

A state is said to be **ergodic** if it is positively recurrent and aperiodic (i.e., the average return time to this state is finite).

Classification of Chain States

Type of States (Summary)

- A state is **recurrent** if it is visited infinitely often.
- A state is **transient** if it may never be visited again after leaving it once.
- A state i is **absorbing** if $p_{ii} = 1$.
- A state i is **periodic** if it returns to itself at regular intervals.
- A state is **ergodic** if it is positively recurrent and aperiodic.

Equivalence Classes

- **Equivalence Class**: A set of states that communicate with each other forms an **equivalence class**:

1. **Closed Class**: Once entered, the system cannot leave.
2. **Transient Classes**: It's possible to leave and never return.
3. **Absorbing Classes**: A closed class with at least one absorbing state, which traps the chain forever.

- **Irreducibility:** A Markov chain is **irreducible** if all states communicate with each other,i.e: It is possible to reach any state from any other state.
- **Ergodicity:** A Markov chain is ergodic if all states are ergodic.

Classification of Chain States

Example

Let a DTMC on the states 0, 1, 2, 3, 4 and 5 with the transition probabilities on one step given by the matrix

$$P = \begin{pmatrix} 0 & 1/3 & 1/3 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Describe the type of its states.

Stationary distribution and limiting probabilities

The transient regime analysis of a Markov chain consists of determining the vector $\pi^{(n)}$ of state probabilities, usually written as

$$\pi_i^{(n)} = \mathbb{P}\left(X_n = i\right)$$

which represents the probability that the chain $\{X_n, n \in N\}$ is in state i after n steps.
The distribution of X_n can be expressed as the row vector:

$$\pi^{(n)} = (\pi_0^{(n)}, \pi_1^{(n)}, \pi_2^{(n)}, \ldots),$$

where for all $i \in S$, $\sum\limits_i \pi_i^{(n)} = 1$.

Stationary distribution and limiting probabilities

To compute $\pi^{(n)}$, one needs either the initial state X_0 or its initial distribution $\pi^{(0)} = (\pi_1^{(0)}, \pi_1^{(n)}, \pi_2^{(0)}, \ldots)$.
By the law of total probability:

$$\mathbb{P}\left(X_n = j\right) = \sum_{i \in S} \mathbb{P}\left(X_0 = i\right) \mathbb{P}\left(X_n = j / X_0 = i\right)$$

$$\pi_j^{(n)} = \sum_{i \in S} \pi_i^{(0)}\, p_{ij}^{(n)}.$$

In matrix form:

$$\pi^{(n)} = \pi^{(0)} P^{(n)} = \pi^{(0)} P^n,$$

and equivalently:

$$\pi^{(n+1)} = \pi^{(n)} P.$$

Stationary distribution and limiting probabilities

Limiting Probabilitie

It is often observed that the distribution $\pi(n)$ converges to a limiting distribution as $n \to \infty$. In that case, this limiting distribution defines the **steady-state regime** of the Markov chain. In practice, it is generally assumed that the steady state is reached after a finite number of transitions.

Definition

Let

$$\pi_j = \lim_{n\to\infty} P_{ij}^n = \lim_{n\to\infty} p_{ij}^{(n)}.$$

π_j represents the **limiting probability** that the chain is in state j , independent of the starting state i.

Stationary Distribution and limiting probabilities

Limiting Distribution

Definition

A Markov chain is said to converge to π if

$$\lim_{n\to\infty} \pi^{(n)} = \pi,$$

independently of the initial distribution $\pi^{(0)}$.
For an n-state DTMC, with states $0, 1, ..., n-1$,

$$\pi = \left(\pi_0, \pi_1, ..., \pi_{n-1}\right),$$

π represents the **limiting distribution** over the state, where $\sum_{i=0}^{n-1} \pi_i = 1.$

Stationary Distribution and limiting probabilities

Stationary Distribution

A Markov chain is **stationary** or **invariant** if its distribution $\pi^{(n)}$ is independent of time n. And independent of the distribution $\pi^{(0)}$ if and only if the matrix P^n converges to a limit P^* whose all rows are identical and equal to π.

Definition

A distribution π on the states of a Markov chain is **invariant** or **stationary** if it satisfies

$$\pi = \pi P. \tag{4}$$

with condition: $\sum_{i=0}^{n-1} \pi_i = 1.$

Stationary Distribution and limiting probabilities

Convergence (Existance and uniqueless)

Theorem

Any finite Markov chain has at least one stationary distribution. However, it may not be unique.

Theorem

If the Markov chain is irreducible and aperiodic, then the stationary distribution exists, regardless of the initial distribution.

Theorem

A finite Markov chain admits a unique stationary distribution iff it has a single recurrent class.

Stationary Distribution and limiting probabilitie

Expected Return Time

Definition

A Markov chain is ergodic if it is irreducible, aperiodic, and has a finite return time to each state.

Theorem

For an irreducible, aperiodic finite-state Markov chain with transition matrix P

$$m_{jj} = \frac{1}{\pi_j} \tag{5}$$

where m_{jj} *is the mean time steps between visits to state* j *and* $\pi_j = \lim_{n\to\infty} (P^n)_{ij} = \lim_{n\to\infty} p_{ij}^{(n)}$.

Stationary Distribution and limiting probabilitie

Remark

- *The existence of stationary distribution implies that the Markov chain will settle into a stable behavior over time.*

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- *The conditions of irreducibility and aperiodicity ensure that the chain can reach any state from any other state and does not cycle periodically.*

Stationary Distribution and limiting probabilitie

Remark

- *The existence of stationary distribution implies that the Markov chain will settle into a stable behavior over time.*
- *The conditions of irreducibility and aperiodicity ensure that the chain can reach any state from any other state and does not cycle periodically.*
- *Thus, for an ergodic Markov chain, the limiting probabilities exist and are independent of the initial conditions, allowing us to calculate the long-term behavior of the system.*

Stationary Distribution and limiting probabilitie

Example (Repair facility problem with cost)

Consider again the Repair facility problem represented by the finite-state DTMC. The help desk is trying to figure out how much to charge me for maintaining my machine. They figure that it costs them 3000 DA every day that my machine is in repair. What will my annual repair bill be?

Stationary Distribution and limiting probabilitie

Example

We have to describe, if possible, the behavior of long-term distributions of the states of three Markov chains, their transition matrices are:

1. $P = \begin{pmatrix} 1/4 & 0 & 3/4 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}$
2. $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
3. $P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$